LECTURE 03 • LOGIC BACKSTAGE

Equivalent forms, identical truth

First, finish the conditional family. Then test how truth survives every rewrite.

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$p \to q$

Sufficient sends; necessary receives

sufficient → necessary

p is sufficient for q

p guarantees q

q is necessary for p

p cannot happen without q

A ticket may be necessary to enter, but owning one may not be sufficient if the gate is closed.

EXERCISE 1

Arrow inspection: accept or repair

e = you enter | **s** = QR code scans | **p** = password is correct | **a** = app opens

Each proposed translation may be right or reversed. Accept or repair it.

1. **"You enter only if the QR code scans." Proposed: $s \to e$** — REPAIR: $e \to s$
2. **"You enter if the QR code scans." Proposed: $s \to e$** — ACCEPT
3. **"A correct password is necessary for the app to open." Proposed: $p \to a$** — REPAIR: $a \to p$
4. **"A correct password is sufficient for the app to open." Proposed: $p \to a$** — ACCEPT

THE CONVERSE TRAP

The converse needs its own evidence

Every dog is a mammal. Every mammal is a dog.

Professor Whiskers has filed an appeal. The dolphin remains unconvinced.

START WITH $p \to q$

Swap, negate, or do both

CONVERSE	INVERSE	CONTRAPOSITIVE
$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
swap	negate	swap + negate

Which relatives always travel together? We will test every row.

EXERCISE 2

Build, then test, the penguin family

> **If an animal is a penguin, then it is a bird.**

Converse	**If it is a bird, then it is a penguin.**	**False: owl.**
Inverse	**If it is not a penguin, then it is not a bird.**	**False: owl again.**
Contrapositive	**If it is not a bird, then it is not a penguin.**	**True.**

IF AND ONLY IF • $p \leftrightarrow q$

Iff is the original plus its converse

$p \to q$ and $q \to p$, locked together.

$p \leftrightarrow q$

A biconditional rewards matching truth values

> $(p \rightarrow q) \land (q \rightarrow p)$

T = T	T ≠ F	F ≠ T	F = F
$p \leftrightarrow q = T$	$p \leftrightarrow q = F$	$p \leftrightarrow q = F$	$p \leftrightarrow q = T$

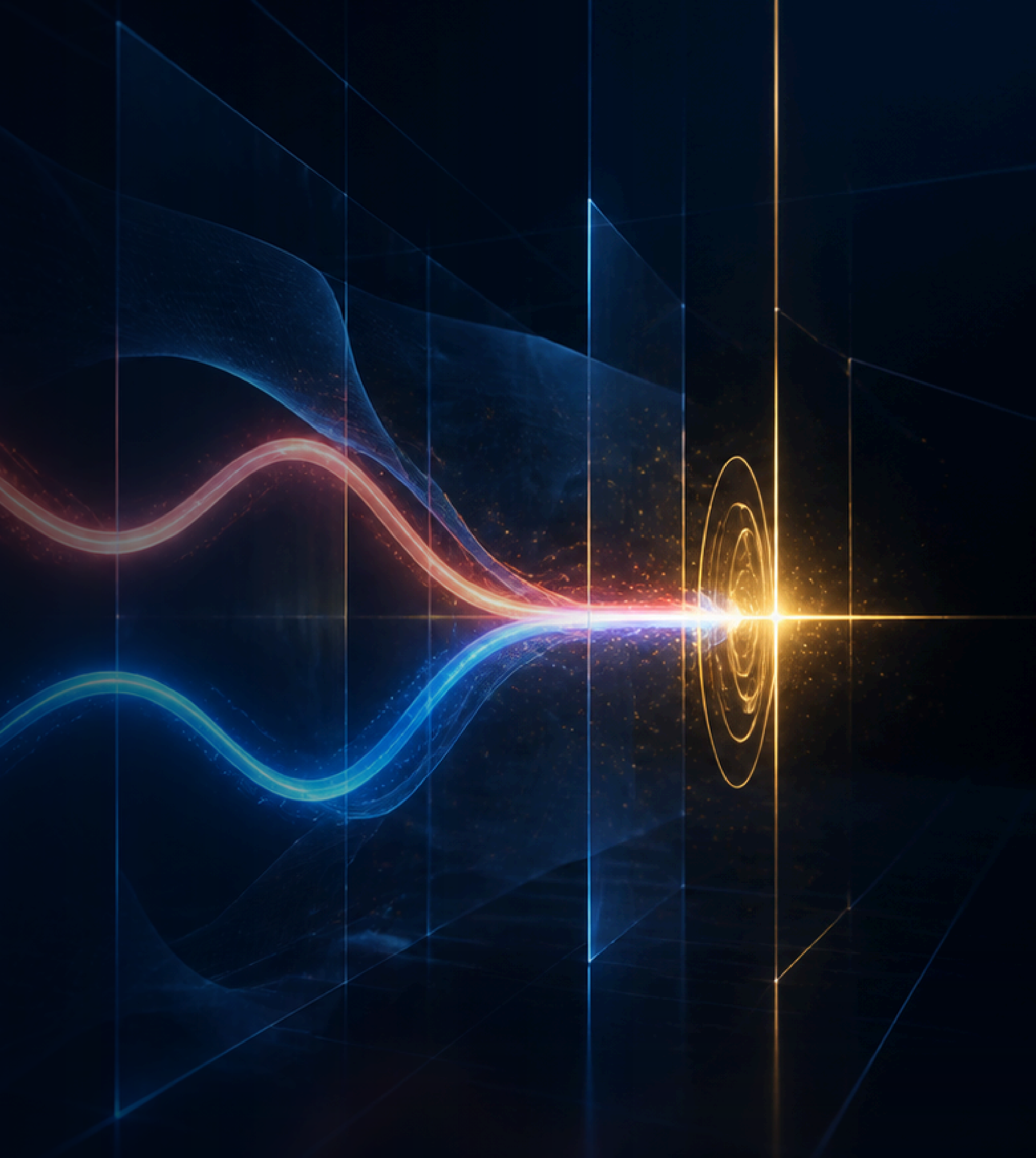
An integer is even iff it is divisible by 2.

Necessary and sufficient—both directions.

EQUIVALENCE OF CONDITIONAL STATEMENTS

Equivalent conditionals: different syntax, identical truth

One counterexample separates them. Every matching row unites them.



Equivalent formulas agree in every possible world

A ≡ B means A and B have the same truth value always

Their final truth-table columns match row for row.

The converse can fail where the original survives

If a file is PNG, then it is an image.	TRUE	Every PNG is an image.
If a file is an image, then it is PNG.	FALSE	A JPEG is a counterexample.
$p \to q$ versus $q \to p$	NOT EQUIVALENT	One world can separate their columns.

Reversing an arrow is a new claim, not a free rewrite.

THE COLUMNS MATCH

A conditional travels with its contrapositive

> $p \rightarrow q \equiv \neg q \rightarrow \neg p$

p	q	$p \rightarrow q$	$\neg q$	$\neg p$	$\neg q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Same four outputs: T, F, T, T.

A NON-OBVIOUS REWRITE

Compilation gives a useful contrapositive

compiled → syntax valid ≡ syntax invalid → did not compile

ORIGINAL	**Successful compilation guarantees valid syntax.**
CONTRA	**Invalid syntax guarantees compilation failed.**
NOT THE CONVERSE	**Valid syntax need not guarantee compilation succeeds.**
WHY	**A missing library can still stop the build.**

Converse and inverse fail together

> $q \to p \equiv \neg p \to \neg q$

p	q	$q \to p$	$\neg p$	$\neg q$	$\neg p \to \neg q$
T	T	T	F	F	T
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	T	T

They may both differ from the original, but they match each other.

A conditional is an OR in disguise

$$p \rightarrow q \equiv \neg p \vee q$$

BAD WORLD	**p is true and q is false.**
THERE	**$\neg p \vee q$ is false.**
ON EARTH	**At least one disjunct is true.**
RESULT	**Both formulas have exactly one false row.**

COMBINE TWO PROMISES

Two routes to one outcome merge with OR

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

ROUTE p	**A staff badge grants entry.**
ROUTE q	**A guest pass grants entry.**
MERGE	**Either credential grants entry.**
FAILURE	**Some valid credential appears, but entry does not.**

Nested promises collapse into one joint trigger

$$p \to (q \to r) \equiv (p \land q) \to r$$

p	**The password is correct.**
q	**The one-time code is correct.**
r	**Access is granted.**
SAME FAILURE	**Both credentials are correct, but access is denied.**

EXERCISE 3

Equivalent—or merely similar?

PAIR A

$p \to q$ versus $q \to p$

F Counterexample: p = T, q = F.

PAIR B

$p \to q$ versus $\neg p \lor q$

T They share the single false row.

PAIR C

$(p \land q) \to r$ versus $p \to (q \to r)$

T They share the joint trigger.

PAIR D

$p \to (q \lor r)$ versus $(p \to q) \lor (p \to r)$

T Both simplify to $\neg p \lor q \lor r$.

TAUTOLOGY · CONTRADICTION

Somethings ignore inputs

All T: tautology. All F: contradiction.

DEFINITION · TAUTOLOGY

A tautology is true in every world

$p \vee \neg p$

No truth assignment makes the final column false.

THREE DIFFERENT SHAPES • ALL TRUE

Tautologies survive every input

$(p \wedge q) \to p$	ALWAYS T	A conjunction cannot hold without p.
$(p \to q) \vee (q \to p)$	ALWAYS T	At least one direction survives.
$(p \to q) \leftrightarrow (\neg q \to \neg p)$	ALWAYS T	A conditional matches its contrapositive.

A tautology is a reusable logical law, not merely a true sentence.

A contradiction is false in every world

$p \land \neg p$

No truth assignment makes the final column true.

THREE DIFFERENT SHAPES • ALL FALSE

Contradictions cannot be rescued by inputs

$p \land \neg p$	ALWAYS F	p cannot have both truth values.
$(p \to q) \land p \land \neg q$	ALWAYS F	It demands the conditional's broken row and its truth.
$(p \leftrightarrow q) \land (p \oplus q)$	ALWAYS F	The first requires a match; the second requires a mismatch.

If a proof reaches a contradiction, its assumptions cannot all be true together.

EXERCISE 4

What survives contact with always?

Question: If t is a tautology and c is a contradiction, then what are the values of $p \wedge t$ and $p \wedge c$?

$p \wedge t$	$p \wedge c$
p	F
AND with true changes nothing.	AND with false is always false.

VALID CHAIN • INVALID REVERSAL

Structure separates a deduction from a guess

SNOW CHAIN

$n \cdot n \to b \cdot b \to c$ — **therefore c**

n: snow in New York • b: snow in Boston • c: Boston schools close.

SERVER REVERSAL

$p \to q \cdot q$ — **therefore p**

The second premise does not reverse the arrow.

Claims → structure → justified conclusion.

The server was innocent. DNS had motive.

Six ideas to carry forward

1
Propositions
A declarative claim in a fixed context has exactly one truth value.

2
Connectives
$\neg$, $\wedge$, $\vee$, and $\oplus$ combine truth values according to precise rules.

3
Truth tables
Test every possible world; matching final columns establish equivalence.

4
De Morgan's laws
Push negation inward, negate every input, and flip $\wedge \leftrightarrow \vee$.

5
Conditionals
$p \rightarrow q$ is false only at $T \rightarrow F$; "only if" names the necessary condition.

6
Formula families
Track converses and contrapositives; recognize tautologies and contradictions.

Use the structure—not the wording—to decide what follows. Next: a systematic equivalence toolkit.